

Day 6: Relational Database & Workflow Automation (Airtable + n8n + Slack)

By: Hafiz Ahmad Raza

Executive Summary

Day 6 focused on replacing flat spreadsheet storage with a robust, relational cloud database using **Airtable**, and connecting it directly with **n8n** and **Slack** to build an automated, real-time Lead Management pipeline.

Architecture & Relational Schema

Created the **MATalogics AI Operations Base** containing 5 core relational tables to manage agency operations:

- **Clients:** Tracks client metadata, status, and linked projects.
- **Projects:** Manages deadlines, assigned team members, and status.
- **Leads:** Captures new inbound leads, service interest, and contact info.
- **AI Agents:** Tracks deployment status, type, and last updated dates.
- **Interns:** Monitors performance scores, task counts, and supervisors.

Automation Workflow (Lead Management Pipeline)

- **Trigger:** Manual / Webhook trigger simulating a new lead submission.
- **Database Action (Airtable Node):** Executes a POST request to insert new lead records directly into the Leads table (Create a Record).
- **Alert System (Slack Node):** Dynamically extracts created record fields (Lead Name, Interested Service, Contact) and dispatches a formatted alert to the #client-updates channel.

Key Engineering Learnings

- **Database vs. Spreadsheet:** Enforced strict data typing (Single Select, Email, Date) in Airtable to prevent API JSON payload mismatches.
- **API Authentication:** Configured Airtable Personal Access Tokens (PAT) with scoped permissions (data.records:write, data.records:read) for secure n8n integration.
- **Dynamic Data Mapping:** Mapped raw JSON outputs from database nodes into downstream Slack notification blocks using n8n expressions (\$json.fields[...]).

Repository Submission Assets

- Day-06/Airtable/: Database schema screenshots (01_Airtable_Base_Created.png).
- Day-06/n8n/: Exported workflow JSON (Task_06_Airtable_Slack.json) and execution log screenshots (03_Workflow1_Lead_Management_Slack.png).